

Business Sales Performance Analytics Report

Future Interns – Data Science & Analytics Task 1

Fellowship Program

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Project: FUTURE_DS_01

Tools Used: Microsoft Excel, Power BI, DAX

1. Executive Summary

This project analyses Superstore business sales data to identify revenue trends, product performance, category-wise profitability, regional profit contribution, and customer-segment sales distribution.

The objective was to transform raw transaction-level sales data into a clear, interactive dashboard that supports business decision-making. The final Power BI dashboard enables stakeholders to filter performance by Year, Region, Category, and Customer Segment.

The analysis shows that the business generated approximately **\$2.30M in total sales** and **\$286.4K in total profit**, resulting in a **12.47% profit margin**. The West region generated the highest profit, Technology was the highest sales-generating category, and the Consumer segment contributed the largest share of revenue.

2. Project Objective

The objective of this project was to answer the following business questions:

- How has sales performance changed over time?
 - Which products generate the highest revenue?
 - Which category contributes the highest sales and profit?
 - Which regions are the most profitable?
 - Which customer segment contributes the largest share of sales?
 - What business actions can improve revenue and profitability?
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3. Dataset Overview

The project uses a Superstore sales dataset containing transaction-level business records.

The dataset includes fields related to:

- Order ID and Order Date
- Ship Date and Ship Mode
- Customer ID and Customer Segment
- City, State, and Region
- Product Name, Category, and Sub-Category
- Sales, Quantity, Discount, and Profit

This dataset represents retail sales transactions across multiple product categories, customer segments, and regions.

4. Data Cleaning and Preparation

Before creating the dashboard, the dataset was reviewed and prepared for analysis.

The following cleaning activities were performed:

- Standardized mixed date formats in the Order Date and Ship Date fields.
 - Created cleaned date columns for accurate time-based analysis.
 - Verified data types for sales, profit, quantity, discount, and date columns.
 - Reviewed blank values, errors, and invalid entries in important fields.
 - Preserved negative profit values because they represent real business losses.
 - Ensured sales, profit, category, region, and product fields were suitable for dashboard analysis.
 - Created DAX measures for Total Sales, Total Profit, Total Orders, Total Quantity, Average Order Value, and Profit Margin.
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5. Key Performance Indicators

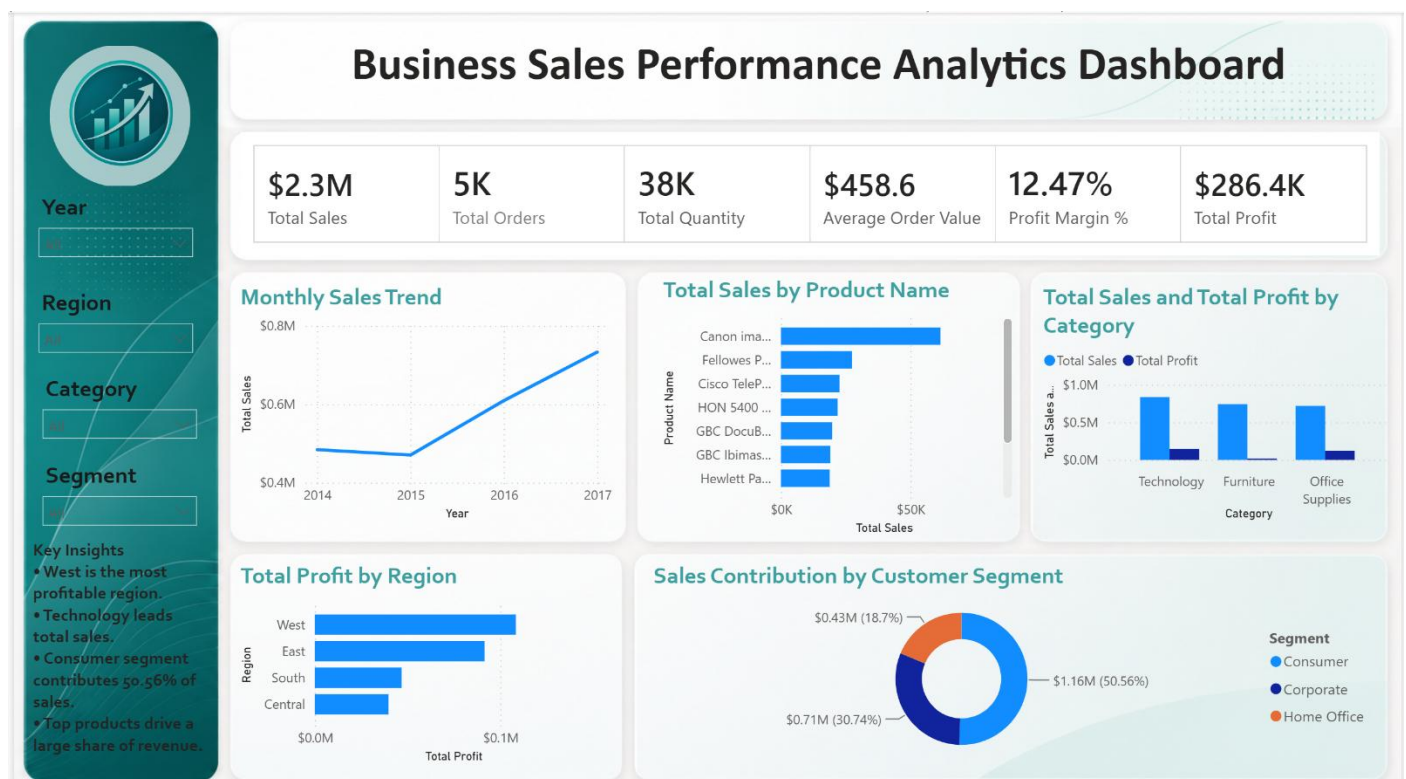
KPI	Value
Total Sales	\$2.30M
Total Profit	\$286.4K
Total Orders	5K
Total Quantity Sold	38K
Average Order Value	\$458.6
Profit Margin	12.47%

The dashboard indicates a healthy overall profit margin of 12.47%. However, revenue growth should be supported by monitoring category-level profit and discount performance.

6. Dashboard Overview

The Power BI dashboard contains the following components:

- KPI cards for overall business performance
- Yearly sales trend analysis
- Top-selling products by revenue
- Sales and profit comparison by category
- Regional profit analysis
- Customer segment sales contribution
- Interactive filters for Year, Region, Category, and Segment



7. Key Findings

7.1 Sales Trend

The sales trend shows a small decline from 2014 to 2015, followed by strong growth in 2016 and 2017. The year 2017 recorded the highest sales performance among the available years.

This indicates that the business experienced stronger revenue growth in the later years of the dataset.

7.2 Category Performance

Technology generated the highest total sales among all categories. Furniture and Office Supplies also made meaningful contributions to sales, but Technology remained the leading category.

Technology should therefore be treated as a priority category for inventory planning, promotions, and product availability.

7.3 Regional Profitability

The West region generated the highest total profit, followed by the East region.

The South and Central regions generated comparatively lower profit. These regions may require localized marketing, pricing reviews, or category-level performance analysis to improve profitability.

7.4 Customer Segment Contribution

The Consumer segment was the largest contributor to total sales.

Customer Segment Sales Contribution

Consumer	\$1.16M — 50.56%
Corporate	\$0.71M — 30.74%
Home Office	\$0.43M — 18.70%

The Consumer segment contributes more than half of total sales, making it the most important audience for customer engagement and sales campaigns.

7.5 Product Performance

A small number of products generated a significant portion of sales revenue. The top product analysis highlights high-value products that should receive attention in stock planning and promotional strategies.

The highest-performing products can be used for:

- Inventory prioritization
- Bundling with lower-performing products
- Promotional campaigns
- Cross-selling opportunities

8. Business Recommendations

Based on the dashboard analysis, the following recommendations are proposed:

8.1 Prioritize High-Performing Categories

Technology should receive priority in marketing campaigns, inventory availability, and customer offers because it generates the highest sales.

8.2 Strengthen Low-Profit Regions

The Central and South regions should be analysed further to identify reasons for lower profit. Possible actions include localized offers, better distribution planning, and region-specific marketing campaigns.

8.3 Focus on Consumer Segment Retention

Since the Consumer segment contributes 50.56% of total sales, the business should create loyalty offers, personalized promotions, and repeat-purchase campaigns for this segment.

8.4 Improve Top Product Availability

Top-selling products should be monitored closely to avoid stockouts. High-demand products can also be bundled with lower-performing items to improve overall sales movement.

8.5 Review Discount Impact on Profitability

Discounts should be monitored carefully, especially for low-margin products. The business should avoid aggressive discounting where it reduces profit without producing meaningful sales growth.

8.6 Use Trend Analysis for Planning

The growth observed in later years suggests that the business should use historical sales patterns for demand forecasting, inventory planning, and campaign scheduling.

9. Conclusion

This project demonstrates how sales data can be transformed into meaningful business insights using Excel and Power BI.

The dashboard provides a clear view of overall revenue, profitability, category performance, customer behaviour, regional contribution, and top products. The analysis shows that Technology, the West region, and the Consumer segment are major drivers of business performance.

The dashboard can support managers and business owners in making more informed decisions related to inventory, promotions, customer targeting, and regional growth strategies.

10. Project Deliverables

The following files are included in the GitHub repository:

- Cleaned Superstore sales dataset
- Power BI dashboard file (.pbix)
- Dashboard screenshots
- Business Sales Analytics Report
- README documentation